
3.3.3 Raw data may require processing. Processed data should be used to plot graphs that illustrate patterns and trends from which appropriate conclusions may be drawn. Scientific knowledge from the AS specification should be used to explain these conclusions.

Candidates should be able to

- process data by carrying out appropriate calculations
- select relevant data to present an effective summary of the results of an investigation and plot this as an appropriate graph conforming to the conventions specified in the Institute of Biology publication, *Biological Nomenclature, Recommendations on Terms, Units and Symbols*, 3rd edition (2000) concerning organisation and presentation of units
- describe, concisely but fully, the trends and patterns in data collected, relating these to specific values, quantities and units
- recognise correlations and causal relationships
- draw valid conclusions, relating explanations to specific aspects of the data collected and applying biological knowledge and understanding from the AS specification.

Practical work carried out in the context of Units 1 and 2 should enable candidates to gain experience of

- using a standard scientific calculator to calculate mean and standard deviation, rate and percentage change
 - plotting data as line graphs, bar charts and histograms
 - plotting data as scatter diagrams and using these to identify correlation.
-